

Tournament Management System Project Proposal

Course Name: Software Engineering

Section: O

Group: 05

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This proposal describes the Tournament Management System aimed at creating efficient management and interaction in gaming tournaments. It indicates the users' functional requirements through use cases and user stories and the basic data model.

Proposal Structure

1. Identify Users & Activities
 2. Use Case Diagram
 4. User Storyboard
 5. Project Links
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1. Identify Users & Activities

The Tournament Management System works with four separate user roles, each with specific tasks and responsibilities:

- Admin

Role: System administrator with full access and control

Key Activities:

- Manage user accounts (activate/deactivate users)
- Manage games catalog (add/edit/delete games)
- Manage teams (create/update/delete team records)

- Manage tournaments (full tournament lifecycle control)
- Generate match schedules for tournaments
- Update match results and standings
- System oversight and maintenance

- Organizer

Role: Tournament creator and manager (subset of Admin permissions)

Key Activities:

- Create and manage their own tournaments
- Set tournament rules and parameters
- Monitor tournament progress
- Coordinate with participants

- Team Captain

Role: Team leader responsible for team management and tournament registration

Key Activities:

- Register team for tournaments
- Manage team roster and member information
- Represent team in tournament communications
- View team's match schedules and results

- Player

Role: Individual participant in tournaments

Key Activities:

- Register for user account
- View available tournaments
- View personal match schedule
- Check tournament standings and results
- Update personal profile information
- Reset/change account password

System Interactions:

- All users connect through the same login system
- Role-based access control determines available features
- Sequential permissions (Admin > Organizer > Team Captain > Player)
- Shared data access for tournaments, teams, and match information

2. Use Case Diagram

The use case diagram illustrates the functional requirements and interactions between different user roles and the tournament management system.

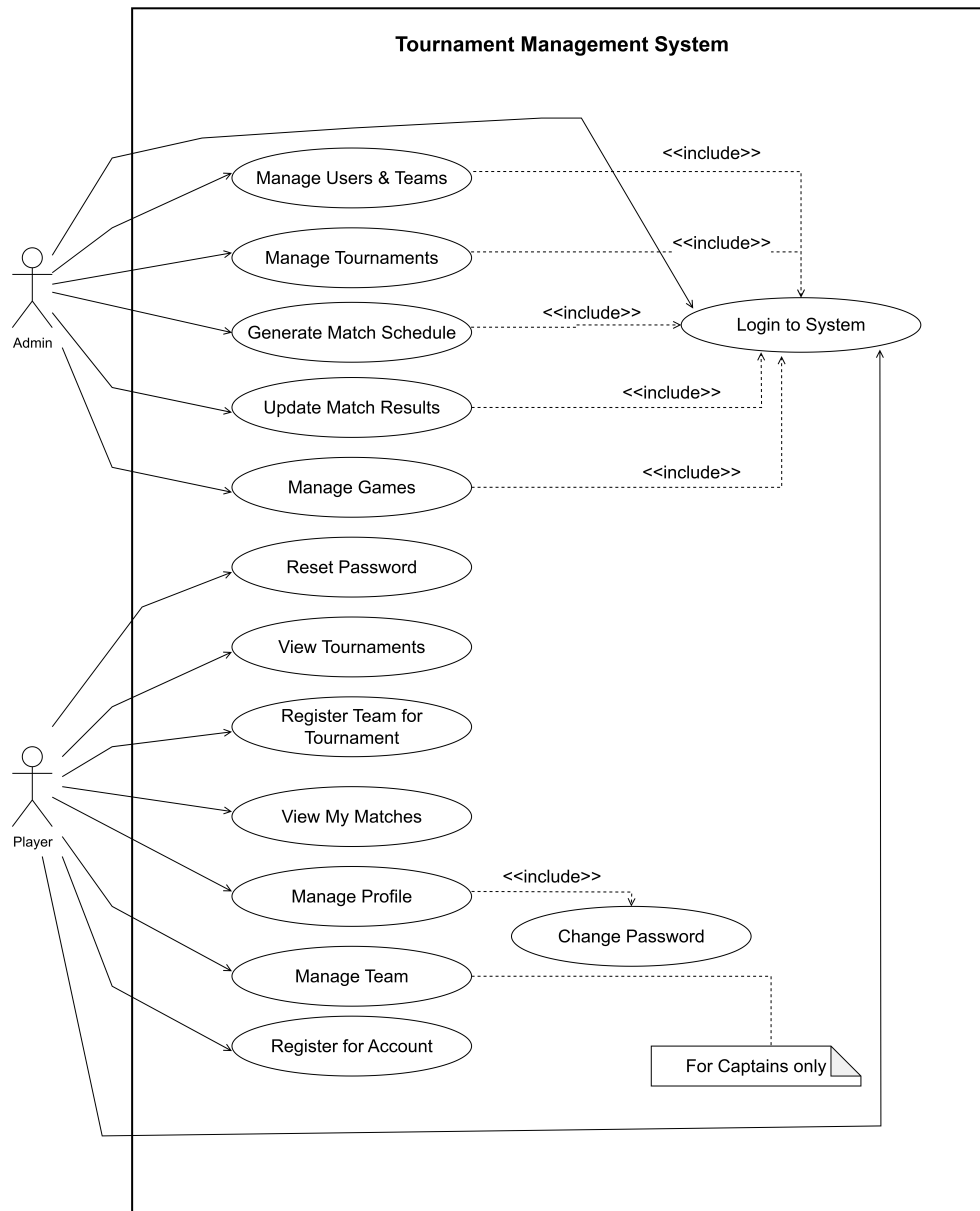


Figure 1: Tournament Management System Use Case Diagram

3. User Storyboard

Below are the main user stories that explain the basic functionality for each user role in the Tournament Management System:

Table 1: Core User Stories for Tournament Management System

User Story	Role	Priority
As an Admin, I want to log in so that I can access the management dashboard.	Admin	Must-Have
As a Player, I want to register an account so that I can join the system.	Player	Must-Have
As an Admin, I want to manage users so that I can control system access.	Admin	Must-Have
As an Admin, I want to manage games so that tournament options are available.	Admin	Must-Have
As an Admin, I want to manage teams so that team information stays organized.	Admin	Must-Have
As an Organizer, I want to create and manage tournaments so that events can run on schedule.	Organizer	Must-Have
As an Admin, I want to generate match schedules so that tournaments have organized fixtures.	Admin	Must-Have
As an Admin, I want to update match results so that standings are accurate.	Admin	Must-Have
As a Team Captain, I want to register my team for tournaments so that we can compete.	Team Captain	Must-Have
As a Player, I want to view available tournaments so that I can decide which to join.	Player	Must-Have
As a Player, I want to view my matches so that I can plan my time.	Player	Must-Have
As a Player, I want to reset my password so that I can regain access to my account.	Player	Must-Have

4. Model Selection:

Based on our project requirements and course materials, we chose **Scrum** as the best suitable process model for the Tournament Management System.

Why Scrum?

- Supports developing needs through iterative sprints, matching our project's need to accept potential modifications.
- Small team size matches with Scrum roles (Product Owner, Scrum Master, Development Team), promoting effective collaboration.
- Focuses on regular delivery of working software, ensuring constant progress and early identification of faults.
- Allows clear communication via daily meetings and sprint reviews, lowering risks.
- Compatible with Agile principles given in course lectures, increasing freedom over inflexible waterfall processes.

Why not Waterfall or other models?

- Waterfall is inflexible and follows fixed criteria, not suitable for our iterative demands.
- Extreme Programming focuses mainly on coding concepts like pair programming; our team's focus is broader.
- Feature Driven Development (FDD) is helpful but better suited for larger teams and well-defined feature lists.

5. Project Links:

1. Public Trello board URL: <https://trello.com/b/9le6XJFY>
2. Public Notion proposal page URL: https://pinto-subway-498.notion.site/Tournament-Management-System-Project-Proposal-252511a6237880b59408d0757b785602?source=copy_link